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$$\begin{aligned}\int_0^{\frac{\pi}{2}} \cos^3 x dx &= \int_0^{\frac{\pi}{2}} (1 - \sin^2 x) d(\sin x) \stackrel{t=\sin x}{=} \int_0^1 (1 - t^2) dt \\ &= \left[t - \frac{t^3}{3} \right]_0^1 = \left(1 - \frac{1}{3} \right) - (0) = \frac{2}{3}\end{aligned}$$

References